

MineTrac AI — Mine Subsidence Monitoring Platform (SIH26)

Complete Hinglish Speech & Voice Note Guide + Technical Explanation PDF

■ Purpose of this Document (Hinglish Guide):

Yeh PDF aapke presentation, voice notes recording, aur team hardware demo ke liye banaya gaya hai. Isme MineTrac Dashboard ke har ek section ka **detailed Hinglish explanation**, exact **word-for-word voice note scripts**, aur **remote deployment guide** (Aap ghar par + Hardware team on-site) fully explain kiya gaya hai.

1. Dashboard Ke Har Component Ka Step-by-Step Explanation (Hinglish)

1. Top Header & Navigation Bar:

- **Logo & Title:** MineTrac BY AKSSAI ProjExcel | SIH26 Coalition.
- **Mine Location Selector:** Dropdown me *Jharia Coalfield — Longwall Panel 4* select karke kisi bhi mine area ka telemetry real-time dekh sakte hain.
- **Audio Siren Toggle:** Audio Siren: MUTED / ACTIVE button. Emergency condition me physical high-decibel siren trigger hota hai.
- **Secondary Tab Bar:** Dashboard, ML Test Bench (3-Tier ML), 90-Day Simulation & Scenarios, aur Hardware & Flashing Guide ke beech switch karne ke liye.

2. Top KPI Summary Metric Cards:

- **Active Mesh Nodes (5/5 Online):** Sabhi 5 surface sensor nodes online hain aur 100% radio signal connectivity de rahe hain.
- **Max Surface Tilt (0.02°):** Surface ground acceleration se calculat hone wala inclination angle. Normal baseline 0.02° hota hai.
- **Max Strain Stress Delta (0.5 $\mu\epsilon$):** Rock mass ke andar digital strain gauge ka tensile stress delta.
- **Subsidence Risk % (2.4% SAFE_STABLE):** AI model dwara calculated overall subsidence collapse probability.

3. Test Scenario Presets Bar (Quick Testing):

- **Case 1 (Normal Operation):** Safe elastic ground, risk 2.4%.
- **Case 2 (Pre-Subsidence Creep):** Slow monotonic tilt ramp (0.05° → 0.35°), risk 45.0%.
- **Case 3 (Active Subsidence):** Ground collapse, rapid tilt (>1.5°), breakwire trip, siren fires, risk 94.2%.
- **Case 4 (Explosive Blasting):** High RMS vibration spike (1.45g) zero tilt ke saath (blast suppressed!).
- **Case 5 (Breakwire Trip):** Physical fissure rupture detection.

4. Underground Longwall Mine Cutaway Schematic:

- **Surface Layer:** Preparation plant, Raw-Coal silo, Elevator shaft, aur 5 surface sensor pins (Slave N1 se Slave N5).
- **Strata Layer:** Overburden sandstone aur shale layers, jo roof fall strain ko upar transmit karte hain.
- **Underground Layer:** *GOB Zone* (collapsed roof void), Hydraulic roof support shearer, Continuous miner section, aur coal pillars.

5. GIS Satellite Map & Master Gateway Early Warning Status:

- **Esri Satellite Map API:** Jharia Coalfield GPS grid par nodes N1-N5, Pits, aur Relay stations map par pinpoint hote hain.
- **Blast Discrimination Filter:** Agar underground dynamite blast hota hai, toh high vibration aane par bhi tilt 0 rehta hai. Gateway ise OPERATIONAL BLAST (FILTERED) bolke false siren ko suppress kar deta hai!

2. Complete Voice Note Speech Scripts (Word-for-Word Hinglish)

■■ Voice Note 1: Top Navigation & Executive Introduction

"Hello everyone! Welcome to MineTrac AI. Yeh humara low-cost, real-time mine subsidence detection, prediction, aur early warning platform hai jo Smart India Hackathon (SIH26) ke problem statement ke liye develop kiya gaya hai. Top bar par aap mine location dropdown dekh sakte hain jo abhi Jharia Coalfield Longwall Panel 4 ko monitor kar raha hai. Yahan active navigation tabs hain—Dashboard, 3-Tier ML Test Bench, 90-Day Simulation Timeline, aur Hardware Flashing Guide. Right side par audio siren control abhi Muted state me hai aur safety director badge gateway connectivity verify karta hai."

■■ Voice Note 2: Top KPI Metric Cards & Test Preset Buttons

"Top KPI cards par focus karte hain: Pehla card hai Active Mesh Nodes, jo dikhata hai ki hamare 5 me se 5 nodes online hain aur 100% radio signal ke saath connected hain. Doosra card hai Max Surface Tilt jo 0.02 degrees par hai—yeh hamara normal safe baseline hai. Teesra card hai Max Strain Stress Delta jo 0.5 microstrain par normal tensile strain dikhata hai. Chautha card hai Subsidence Risk Percentage jo 2.4% par Safe Stable status dikha raha hai. Directly neeche 5 quick test preset buttons hain jisse hum Normal, Pre-Subsidence, Active Collapse, aur Dynamite Blasting scenarios ko 1-click me test kar sakte hain."

■■ Voice Note 3: Underground Mine Cutaway Schematic

"Yeh middle section me hamara anatomical underground longwall mine cross-section visualizer hai. Top surface par hamare preparation plant, coal silo, elevator shaft head, aur 1500m × 200m surface grid par Slave N1 se Slave N5 tak ke 5 sensor pins mounted hain. Neeche overburden sandstone aur shale strata layers hain jo underground ground deformation ko surface tak transmit karti hain. Sabse neeche underground area me GOB Zone—yaani collapsed roof void—hydraulic roof support shearer, aur coal pillars dikhaye gaye hain. Jaise hi underground caving start hoti hai, surface pins aur GOB zone border ka color real-time dynamic update hota hai."

■■ Voice Note 4: GIS Satellite Map & Node Mesh Overlay

"Neeche left side par hamara Surface Panel GIS Map hai jo High-Definition Esri World Imagery Satellite Tiles use karta hai. Yeh map Jharia Coalfield ke actual GPS coordinates par surface sensor nodes ko pinpoint karta hai. Node N3 center max-stress zone par positioned hai. Marker par click karne se node ka real-time tilt, strain, vibration, battery, aur TinyML isolation forest status popup me khul jaata hai. Ground risk ke aadhar par color dots green, yellow, red, ya purple me transform ho jaate hain."

■■ Voice Note 5: Gateway Early Warning Status & Blast Filter

"Bottom right side par hamara Master Gateway Early Warning Status panel hai jo 0.0 se 1.0 ke scale par live subsidence risk score dikhata hai. Jab underground dynamite blasting hoti hai, toh traditional sensors false alarm trigger kar dete hain aur mining ko faltu me rokna padta hai. Lekin hamara Stage 2 Gateway Spatial Correlator detect karta hai ki high vibration ke saath TILT 0 hai. Ise yeh OPERATIONAL BLAST (FILTERED) classify karke physical siren ko suppress kar deta hai, jisse false alarm rukta hai aur mining bina kisi rukaawat ke chalti rehti hai!"

■■ Voice Note 6: Interactive 3-Tier ML Test Bench & Sliders

"Ab hum ML Test Bench tab par hain. Yahan Node N3 ke 8 physical sensor sliders—Tilt Mean, Tilt Rate, Strain Delta, Vibration RMS, Peak, Frequency, aur Crack Breakwire—ko live manipulate kar sakte hain. Right side par hamara 3-Tier ML Model inference result dikhata hai:

1. Stage 1 Edge TinyML (ESP32-S3) Isolation Forest C model ko 86.4 microseconds me execute karta hai.
2. Stage 2 Gateway Correlator adjacent nodes ka spatial correlation aur blast suppression evaluate karta hai.
3. Stage 3 Cloud Multi-Output LSTM (TensorFlow Keras) 24-hour sequence window se surface displacement millimeters me aur severity class predict karta hai."

■■ Voice Note 7: 90-Day Indian Longwall Mining Timeline (Phase 1 - 4)

"Finally, hamare 90-Day Simulation tab me hum Longwall Extraction ke pure 4 geomechanical phases ko 50 time-series

windows me play back karte hain:

- *Phase 1 (Normal, Days 0-60): Flat baseline ground (Tilt 0.02°, Risk 2.4%).*
- *Phase 2 (Pre-Subsidence Creep, Days 60-75): Overburden micro-cracking, steady tilt ramp (0.05° → 0.37°), rising strain (14.5 µε), triggering early warning at 45% risk.*
- *Phase 3 (Active Collapse, Days 75-85): Main roof falls into GOB void, rapid tilt acceleration (>1.5°), strain spike (58 µε), breakwire flip, triggering 94.2% risk & siren alert! ■*
- *Phase 4 (Dynamite Blast Event): Explosive vibration spikes to 1.45g while tilt stays ZERO (0.02°), demonstrating blast false-alarm suppression."*

3. Remote Deployment Guide in Hinglish (Ghar + On-Site Team)

■ AAP (Ghar Par - Remote Developer):

1. Github repository https://github.com/Monaal5/mine_subsidence_ml.git par code updated hai.
2. **Render.com** par New Web Service banakar GitHub connect karein. Docker select karke Deploy par click karein.
3. 2 minute me Render आपको public link dega: <https://mine-subsidence-app.onrender.com>.
4. Modifies RAM usage under 200MB & dynamic PORT binding fixes free-tier memory issues!

■ HARDWARE TEAM (On-Site Location):

1. **ESP32-S3 Sensor Nodes Flashing:** ESP32-S3 ko USB se laptop me connect karein. Arduino IDE me `esp32_firmware/main.ino` kholein. Board *ESP32S3 Dev Module* aur COM port select karke Upload (Arrow) button dabayein.
2. **Local Gateway Starter:** Gateway PC par `run_local_gateway.bat` ko double click karein. Server 1-click me port 8050 par chalu ho jaayega.